SARTHI / AUTOPATCH

AI-Powered Self-Healing API Middleware | Executive System Architecture & Whitepaper

1. Executive Summary & Problem Statement

Modern cloud microservices rely heavily on third-party SaaS and financial APIs (Stripe, Shopify, Twilio, Salesforce). When upstream vendors introduce unannounced schema changes (e.g. renaming customer_id to customerId), downstream consumers crash with fatal KeyError and HTTP 500 exceptions. Traditional resolution takes an average of **3.5 to 4.2 hours** of engineer on-call toil, causing customer-facing outages and financial losses. **Sarthi** is an autonomous reverse proxy middleware that reduces Mean Time to Resolution (MTTR) to **under 2 minutes**.

2. Operational Benchmarks & ROI Telemetry

Metric / Dimension	Legacy / Manual Process	Sarthi Autonomous Engine
Mean Time to Detect (MTTD)	15 to 45 minutes (User alerts)	< 180 milliseconds (Proxy intercept)
Mean Time to Resolve (MTTR)	3.5 to 4.2 hours	< 2 minutes (End-to-end recovery)
Repair Cost per Incident	$450.00 (Senior SRE hours)	$0.042 (LLM Token compute)
Security Validation	Manual code review in PR	Python AST Parse + Non-Root Docker Sandbox
Production Deployment Risk	High (Hotfix rushed at 2 AM)	Governed (Two-stage RBAC human review)

3. Five-Stage Autonomous Self-Healing Lifecycle

- **Stage 1 (Reverse Proxy Interceptor):** Sits as a transparent ASGI reverse proxy, inspecting responses against Pydantic OpenAPI schemas in <180ms without buffering bottlenecks.
- **Stage 2 (Incident Deduplication):** Generates deterministic SHA-256 fingerprints on structural diffs, collapsing thousands of concurrent failures into a single root-cause incident.
- **Stage 3 (Qdrant Vector RAG):** Retrieves official vendor migration guides and OpenAPI 3.1 specifications to ground adapter synthesis and eliminate LLM hallucinations.
- **Stage 4 (AST & Docker Sandbox Verification):** Enforces zero-trust AST scanning (disallowing eval/subprocess/network) and executes 4/4 pytest regression assertions in a non-root, read-only Docker container.
- **Stage 5 (Governed Deployment Gate):** Enforces two-stage RBAC approval (Admin/Reviewer) and exports an atomic GitHub PR with a 12-section root cause analysis.

4. Enterprise Security & Compliance Perimeters

Sarthi adheres to strict enterprise security standards: (1) **Zero Arbitrary Code Execution** via Python AST node parsing; (2) **Non-Root Ephemeral Sandbox** with network isolation (--net=none) and strict cgroups (0.5 CPU, 256MB RAM); (3) **Append-Only Audit Ledger** recording all user roles, timestamps, and payload deltas to satisfy SOC 2 Type II non-repudiation.